ASSIGNMENT 1 – CO1 | TASK 7

Procedural versus Object-Oriented Design

Scenario: Namma Metro Smart-Card Fare System

Objective / explanation

This program compares a free-function approach with a SmartCard class. Both are O(1) per journey, but encapsulation prevents outside code from directly changing the balance.

C++ source code

```
#include &lt;iostream&gt;
using namespace std;

bool canTravelProcedural(long long balance, long long fare) {
    return fare &gt;= 0 &amp;&amp; balance &gt;= fare &amp;&amp; balance - fare &gt;= 5000;
}
bool deductProcedural(long long&amp; balance, long long fare) {
    if (!canTravelProcedural(balance, fare)) return false;
    balance -= fare;
    return true;
}
class SmartCard {
    long long balancePaise;
public:
    explicit SmartCard(long long initial) : balancePaise(initial) {}
    bool deduct(long long fare) {
        if (fare &lt; 0 || balancePaise &lt; fare ||
            balancePaise - fare &lt; 5000) return false;
        balancePaise -= fare;
        return true;
    }
    long long balance() const { return balancePaise; }
};
int main() {
    long long proceduralBalance = 10000;
    cout &lt;&lt; "Procedural success: "
         &lt;&lt; deductProcedural(proceduralBalance, 2000) &lt;&lt; '\n';
    SmartCard card(10000);
    cout &lt;&lt; "OOP success: " &lt;&lt; card.deduct(2000) &lt;&lt; '\n';
    cout &lt;&lt; "OOP balance: " &lt;&lt; card.balance() &lt;&lt; '\n';
}
```

Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

Actual captured program output

```
Procedural success: 1
OOP success: 1
OOP balance: 8000
```

Output screenshot

Analysis / complexity

Both implementations take O(1) time and O(1) space per card. A public balance would allow other code to violate the minimum-balance rule; private state protects the invariant.

Evidence note: Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.